

Counting by Leading Meet in Free Distributive Lattices

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Abstract

The free distributive lattice $FD(n)$ has $M(n)$ elements, where $M(n)$ is the n th Dedekind number. While much work has focused on computing these totals, comparatively little attention has been given to the internal distribution of elements within $FD(n)$.

In this paper we enumerate the elements of $FD(n)$ for $3 \leq n \leq 7$ and organize them according to the leading meet in their canonical join-of-meets representation. The resulting tables give, for every meet of size k , the number of elements whose canonical representation begins with that meet.

These data reveal several regular patterns in the combinatorial structure of $FD(n)$. In particular, we establish a geometric law governing meets of size $n - 1$, a recurrence relation for the number of elements whose leading generator is A_i , and a doubling phenomenon for meets of size $n - 2$ when a new generator is introduced.

The primary purpose of this work is to describe the distributions revealed by these computations. The observed patterns and their proofs illustrate some of the combinatorial patterns reflected in the data.

1 Introduction

Free distributive lattices occupy a central place in lattice theory and in the study of monotone Boolean functions. The free distributive lattice $FD(n)$ on n generators has $M(n)$ elements, where $M(n)$ is the n th Dedekind number. These numbers grow extremely rapidly, and considerable effort has been devoted to determining their values.

Most work on Dedekind numbers focuses on the total number of elements of $FD(n)$, often exploiting the symmetries of the Boolean lattice to simplify the enumeration. Much less attention has been given to the internal distribution of these elements when they are organized according to natural parameters.

In this paper we study one such parameter. Every element of $FD(n)$ admits a canonical representation as a join of meets of generators in which no meet term contains another. Ordering the meet terms lexicographically produces a distinguished first term, which we call the *leading meet*. This provides a natural way to partition the elements of $FD(n)$ according to the meet that appears first in their canonical representation. We emphasize that this lexicographic ordering and the associated notion of leading meet are introduced here as a convenient framework for organizing computations, and are not intended as canonical features of the lattice.

Unlike most approaches to Dedekind numbers, which attempt to exploit the symmetries of the Boolean lattice to reduce the computational burden, our analysis deliberately fixes an asymmetric ordering of the generators and studies the behavior that emerges relative to that ordering. The resulting distributions therefore do not reflect the intrinsic symmetries of $FD(n)$, but rather the combinatorial structure that appears when the elements are organized lexicographically by their canonical meet terms. This perspective exposes several regularities in the internal organization of the lattice that are not immediately visible from the symmetric viewpoint typically used in Dedekind number computations.

Using an enumeration program, we computed the complete distribution of elements by leading meet for all values $3 \leq n \leq 7$.

For each meet of generators we record the number of elements of $FD(n)$ whose canonical representation begins with that meet.

The primary purpose of this paper is to present the computed distributions and to describe several regularities that emerge from them. The resulting tables describe how the elements of $FD(n)$ are organized relative to the lexicographic ordering of meets.

Inspection of the data reveals several regular patterns. In particular:

- The counts associated with meets of size $n - 1$ follow a simple geometric law.
- The numbers of elements whose leading meet is the single generator A_i satisfy a natural recurrence relation.
- The counts associated with meets of size $n - 2$ exhibit a regular doubling phenomenon when a new generator is introduced.

These patterns are not immediately evident from the definition of the lattice itself. We therefore give proofs explaining why they occur. The results should be viewed primarily as examples of the behavior reflected in the data.

The remainder of the paper is organized as follows. Section 2 reviews the canonical antichain representation of elements of $FD(n)$ and describes the ordering used in our computations. Sections 3–5 analyze the patterns described above. In the final section we present representative portions of the computed tables. The complete datasets for $3 \leq n \leq 7$, together with the source code used to generate them, are provided as supplementary material.

2 Representation and Enumeration

To study the internal structure of $FD(n)$ we enumerate its elements using the standard representation of lattice elements as antichains in the Boolean lattice. This representation provides both a natural combinatorial description and a convenient basis for computation.

2.1 Antichain Representation

Every element of the free distributive lattice $FD(n)$ admits a canonical representation as a join of meets of generators,

$$(A_{i_1} \wedge \cdots \wedge A_{i_k}) \vee (A_{j_1} \wedge \cdots \wedge A_{j_\ell}) \vee \cdots .$$

In this representation no meet term contains another. If one meet contains another then the larger term is redundant, since

$$(A_{i_1} \wedge \cdots \wedge A_{i_k}) \vee (A_{i_1} \wedge \cdots \wedge A_{i_k} \wedge A_j) = A_{i_1} \wedge \cdots \wedge A_{i_k} .$$

Each meet term therefore corresponds to a subset of the generators $\{A_1, \dots, A_n\}$. The requirement that no meet contain another is equivalent to requiring that these subsets form an antichain in the Boolean lattice of all subsets of $\{A_1, \dots, A_n\}$.

Thus elements of $FD(n)$ are in bijection with antichains of the Boolean lattice $2^{\{A_1, \dots, A_n\}}$. Under this correspondence the number of elements of $FD(n)$ is the number of antichains in this lattice, namely the Dedekind number $M(n)$.

2.2 Bitmask Encoding

For computational purposes we represent each meet by a bitmask indicating which generators appear in the meet.

Specifically, a meet

$$A_{i_1} \wedge A_{i_2} \wedge \cdots \wedge A_{i_k}$$

is encoded by an n -bit integer whose j th bit is 1 if A_j appears in the meet and 0 otherwise.

For example, in $FD(4)$ the meet

$$A_1 \wedge A_3$$

is represented by the bitmask

$$0101.$$

Subset relations between meets can therefore be tested efficiently using bit operations.

2.3 Ordering of Meets

We order meets in two stages.

First, meets are grouped by size, that is, by the number of generators appearing in the meet. Meets containing fewer generators precede those containing more generators.

Within each size class, meets are ordered according to the numerical value of their bitmask representation. Thus the ordering is determined entirely by the binary encoding described above.

For example, in $FD(4)$ the meets of size two appear in the order

$$A_1 \wedge A_2, A_1 \wedge A_3, A_2 \wedge A_3, A_1 \wedge A_4, A_2 \wedge A_4, A_3 \wedge A_4.$$

Under this ordering every element of $FD(n)$ written in canonical join-of-meets form has a well-defined first meet term. We call this term the *leading meet*. Grouping elements according to their leading meet partitions the elements of $FD(n)$ into disjoint classes.

Lemma 1. *Let meets be ordered by size and then by the numerical value of their bitmask representation. When a new generator A_{n+1} is introduced, meets not containing A_{n+1} retain their relative order.*

Among meets of a fixed size k , the meets containing A_{n+1} form a suffix of the ordering. Moreover, their relative order agrees with the ordering of the corresponding $(k-1)$ -generator meets obtained by removing A_{n+1} .

Proof. In the bitmask representation, adjoining A_{n+1} corresponds to adding a new highest bit. If a meet has bitmask b , then the meet obtained by adjoining A_{n+1} has bitmask $2^n + b$.

Thus all masks containing the new bit are larger than those that do not contain it, and within this collection the ordering is determined by the values of the original masks b . $\square$

2.4 Enumeration Procedure

The enumeration proceeds by constructing antichains whose smallest element in the above ordering is fixed.

For n generators there are $2^n - 1$ nonempty meets of generators. Each such meet may serve as the leading meet of a canonical representation. For each candidate leading meet T we enumerate all antichains whose smallest element is T .

Two meets are comparable if the set of generators appearing in one contains the set appearing in the other. Prior to the search we precompute, for each meet, the collection of other meets comparable with it. These relations are stored as bitmasks, allowing rapid elimination of comparable terms during the enumeration.

For a fixed leading meet T we perform a backtracking search that constructs joins by adding meets larger than T in the ordering.

The search begins with the singleton antichain $\{T\}$. At each step we maintain a set of candidate meets that are larger than all previously chosen terms and incomparable with them. Selecting a new meet removes from consideration every meet comparable with it. Each resulting set of mutually incomparable meets corresponds to one element of $FD(n)$.

Because the leading meet is fixed, the enumeration for different choices of T can be performed independently. The counts obtained for each leading meet are then accumulated to produce the distributions studied in the following sections.

3 Representative Data

To provide a concrete view of the objects under consideration, we present the complete data for $FD(4)$ and $FD(5)$. These examples illustrate the combinatorial structure that underlies the patterns analyzed in the subsequent sections. For readability in tables we represent $A_1, A_2, \dots$ by $A, B, \dots$.

For $n = 4$, the elements of $FD(4)$, grouped by leading meet, are as follows:

Meet	Count
A	19
B	14
C	11
D	9
$A \wedge B$	41
$A \wedge C$	23
$B \wedge C$	13
$A \wedge D$	10
$B \wedge D$	6
$C \wedge D$	4
$A \wedge B \wedge C$	8
$A \wedge B \wedge D$	4
$A \wedge C \wedge D$	2
$B \wedge C \wedge D$	1

For $n = 5$, we obtain:

Meet	Count
A	167
B	148
C	134
D	123
E	114
$A \wedge B$	1,556
$A \wedge C$	1,027
$B \wedge C$	687
$A \wedge D$	585
$B \wedge D$	410
$C \wedge D$	309
$A \wedge E$	315
$B \wedge E$	230
$C \wedge E$	179

Meet	Count
$D \wedge E$	145
$A \wedge B \wedge C$	621
$A \wedge B \wedge D$	331
$A \wedge C \wedge D$	177
$B \wedge C \wedge D$	95
$A \wedge B \wedge E$	82
$A \wedge C \wedge E$	46
$B \wedge C \wedge E$	26
$A \wedge D \wedge E$	20
$B \wedge D \wedge E$	12
$C \wedge D \wedge E$	8
$A \wedge B \wedge C \wedge D$	16
$A \wedge B \wedge C \wedge E$	8
$A \wedge B \wedge D \wedge E$	4
$A \wedge C \wedge D \wedge E$	2
$B \wedge C \wedge D \wedge E$	1

Note: The counts in each table sum to $M(n) - 3$, as they exclude the bottom element $\mathbf{0}$, the top element $\mathbf{1}$, and the meet of all generators $A_1 \wedge \cdots \wedge A_n$. The latter is comparable to every other meet and therefore cannot appear as a leading meet in any antichain.

Several patterns are already apparent in these tables. The elements are naturally grouped by their leading meet, and within each group the counts decrease in a regular fashion as additional generators are included.

More significantly, the ordering exhibits the suffix structure described in Section 2: when a new generator is introduced, the meets obtained by adjoining it to earlier meets appear as a contiguous block at the end of the ordering. This behavior is visible in the transition from $n = 4$ to $n = 5$, where all meets involving E occur after those involving only A, B, C, D , and within this block they preserve the relative order of the earlier sequence.

These examples also begin to reveal systematic numerical patterns. In particular, the counts associated with larger meets display regular growth across successive values of n , a phenomenon that will be examined in detail in the following sections.

Complete data for $n = 6$ and $n = 7$, which exhibit the same behavior on a larger scale, are provided in the supplementary material.

4 Geometric Sequence for $(n - 1)$ -Meets

A clear regularity appears among meets of size $n - 1$. Extracting the $(n - 1)$ -meets for $n = 5$ from the table above, we obtain:

Meet	Count
$A \wedge B \wedge C \wedge D$	16
$A \wedge B \wedge C \wedge E$	8
$A \wedge B \wedge D \wedge E$	4
$A \wedge C \wedge D \wedge E$	2
$B \wedge C \wedge D \wedge E$	1

These values form a geometric sequence with ratio $\frac{1}{2}$.

This behavior reflects the ordering described in Section 2. The $(n-1)$ -meets occur as a final segment of the lexicographic ordering, and successive meets correspond to systematically replacing earlier generators. At each step, the number of admissible extensions is reduced by a factor of 2.

This observation suggests the following general statement.

Proposition 1. *Let $FD(n)$ be the free distributive lattice on generators $A_1, \dots, A_n$.*

Under the canonical ordering, the number of elements whose leading meet is the i th meet of size $n-1$ is

$$2^{n-i}.$$

Consequently the counts form the geometric sequence

$$2^{n-1}, 2^{n-2}, \dots, 2^0.$$

Proof. A meet of size $n-1$ corresponds to a subset of the generators omitting exactly one generator. These meets are mutually incomparable, so any subset of them forms an antichain and therefore determines a valid element of $FD(n)$.

Fix the i th such meet in the canonical ordering and require it to be the leading meet. Since the leading meet is the smallest in the ordering, any additional meet in the join representation must occur after it.

There are $n-i$ such later meets, since there are n meets of size $n-1$ in total. An element with this leading meet is therefore obtained by choosing an arbitrary subset of these $n-i$ meets, and each choice produces a distinct element.

Hence the number of such elements is

$$\sum_{k=0}^{n-i} \binom{n-i}{k} = 2^{n-i}.$$

□

5 Recurrences for Generator-Leading Elements

Another pattern visible in the computed tables concerns the number of elements whose leading meet is a single generator.

The following table shows representative values obtained from the enumeration.

Table 1: Counts of elements with leading generator A_i for $n = 3$ through 7

n	A	B	C	D	E	F	G
3	5	3	2				
4	19	14	11	9			
5	167	148	134	123	114		
6	7580	7413	7265	7131	7008	6894	
7	7,828,353	7,820,773	7,813,360	7,806,095	7,798,964	7,791,956	7,785,062

Inspection of these values suggests a simple recurrence relation among the counts. The next proposition explains this phenomenon.

Let $G_i(n)$ denote the set of elements of $FD(n)$ whose leading generator is A_i , and write $g_i(n) = |G_i(n)|$.

Proposition 2. *For the first generator A_1 ,*

$$g_1(n) = M(n-1) - 1.$$

Proof. We construct all elements of $FD(n)$ whose leading generator is A_1 directly from the smaller lattice $FD(n-1)$.

Recall that in our lexicographic order, the generators are ordered $A_1 < A_2 < \dots < A_n$. Thus A_1 is the smallest possible leading generator.

For the purposes of this argument only, we identify $FD(n-1)$ with the free distributive lattice on generators $A_2, \dots, A_n$, so that A_1 may be treated as a newly adjoined generator. Every element x of this lattice, except the bottom element $\mathbf{0}$ and the top element $\mathbf{1}$, can be joined with A_1 to produce a new element

$$A_1 \vee x$$

in $FD(n)$. In this join, A_1 is the leading generator because it is smaller than every generator appearing in x . Moreover, distinct elements x give distinct joins.

In addition, the singleton element A_1 itself also has A_1 as its leading generator.

No other elements of $FD(n)$ have A_1 as their leading generator. Any such element must include A_1 in its join representation, and cannot include any smaller generator. Therefore it must be either A_1 itself or a join of the form $A_1 \vee x$, where $x \in FD(n-1)$.

If $x = \mathbf{0}$, this yields A_1 , which has already been counted. If $x = \mathbf{1}$, this yields $A_1 \vee \mathbf{1} = \mathbf{1}$, which has no leading generator. These cases are therefore excluded.

Therefore the elements with leader A_1 are exactly:

- A_1 , and
- $A_1 \vee x$ for each $x \in FD(n-1) \setminus \{\mathbf{0}, \mathbf{1}\}$.

The number of such x is $M(n-1) - 2$, giving a total of

$$1 + (M(n-1) - 2) = M(n-1) - 1.$$

□

Proposition 3. For $1 \leq i < n$,

$$g_i(n) = g_i(n-1) + g_{i+1}(n).$$

Proof. First consider the map

$$\phi(x) = x \vee A_n,$$

defined for $x \in FD(n-1)$ whose leading generator is A_i .

To see that ϕ is injective, suppose $\phi(x_1) = \phi(x_2)$. Then

$$x_1 \vee A_n = x_2 \vee A_n.$$

Since neither x_1 nor x_2 contains A_n , the canonical join representation of each side consists of the join representation of x_i together with the meet A_n . By uniqueness of canonical representations we must therefore have $x_1 = x_2$.

The image of ϕ consists precisely of those elements of $G_i(n)$ whose canonical form contains A_n . Since ϕ is injective and its domain contains $g_i(n-1)$ elements, its image contributes $g_i(n-1)$ elements to $G_i(n)$.

Let $G'_i(n)$ denote the remaining elements of $G_i(n)$, namely those whose canonical form does not contain A_n .

Now consider the cyclic permutation of the generators defined by

$$\pi(A_j) = A_{j+1} \quad (1 \leq j < n), \quad \pi(A_n) = A_1.$$

Since $FD(n)$ is free on the generators $A_1, \dots, A_n$, any permutation of the generators extends uniquely to a lattice automorphism.

If $x \in G'_i(n)$, then the canonical form of x contains A_i as its leading generator and does not contain A_n . Applying π replaces each generator A_j by A_{j+1} . Since A_n does not occur in x , no generator is mapped to A_1 , and the leading generator becomes A_{i+1} . Thus

$$\pi(G'_i(n)) \subseteq G_{i+1}(n).$$

Conversely, let $y \in G_{i+1}(n)$. Applying π^{-1} sends A_{i+1} to A_i , so the leading generator of $\pi^{-1}(y)$ is A_i . Moreover, since y contains no occurrence of A_1 , the element $\pi^{-1}(y)$ contains no occurrence of A_n . Hence $\pi^{-1}(y) \in G'_i(n)$, showing that π maps $G'_i(n)$ bijectively onto $G_{i+1}(n)$.

Therefore

$$g_i(n) = |G_i(n)| = |\phi(FD(n-1))| + |G'_i(n)| = g_i(n-1) + g_{i+1}(n).$$

□

6 Diagonal Doubling for $(n - 2)$ -Meets

One of the most striking regularities in the data appears among meets of size $n - 2$. The following table shows representative values illustrating this behavior for the transitions $n=4 \rightarrow 5$ and $n=5 \rightarrow 6$.

Table 2: Doubling pattern for meets of size $n - 2$

$n = 3$ (1-meets)		$n = 4$ (2-meets)		$n = 5$ (3-meets)		$n = 6$ (4-meets)	
Meet	Count	Meet	Count	Meet	Count	Meet	Count
A	5	$A \wedge B$	41	$A \wedge B \wedge C$	621	$A \wedge B \wedge C \wedge D$	18,641
B	3	$A \wedge C$	23	$A \wedge B \wedge D$	331	$A \wedge B \wedge C \wedge E$	9,631
C	2	$B \wedge C$	13	$A \wedge C \wedge D$	177	$A \wedge B \wedge D \wedge E$	4,981
		$A \wedge D$	10	$B \wedge C \wedge D$	95	$A \wedge C \wedge D \wedge E$	2,579
		$B \wedge D$	6	$A \wedge B \wedge E$	82	$B \wedge C \wedge D \wedge E$	1,337
		$C \wedge D$	4	$A \wedge C \wedge E$	46	$A \wedge B \wedge C \wedge F$	1,242
				$B \wedge C \wedge E$	26	$A \wedge B \wedge D \wedge F$	662
				$A \wedge D \wedge E$	20	$A \wedge C \wedge D \wedge F$	354
				$B \wedge D \wedge E$	12	$B \wedge C \wedge D \wedge F$	190
				$C \wedge D \wedge E$	8	$A \wedge B \wedge E \wedge F$	164
						$A \wedge C \wedge E \wedge F$	92
						$B \wedge C \wedge E \wedge F$	52
						$A \wedge D \wedge E \wedge F$	40
						$B \wedge D \wedge E \wedge F$	24
						$C \wedge D \wedge E \wedge F$	16

Note: Bolded counts highlight the suffix doubling—each bolded entry in the $n=5$ column is twice the corresponding entry in the $n=4$ column, and the bolded entries in the $n=6$ column are twice those in the $n=5$ column.

The case $n = 3$ appears here as the boundary case, where $(n - 2)$ -meets are simply generators. The diagonal pattern visible in Table 2 therefore begins at $n = 3$, and the doubling behavior described below holds for all $n \geq 3$.

Inspection of the table shows that when a new generator is adjoined, the counts associated with these meets double along the diagonal. This pattern reflects a feature of the ordering described in Section 2. When a new generator is adjoined, the meets obtained by adjoining it to earlier meets appear as a suffix in the lexicographic ordering, and the $(n - 2)$ -meets of $FD(n + 1)$ arise in this way from those of $FD(n)$.

We illustrate this suffix structure explicitly for the transition from $n = 4$ to $n = 5$.

For $n = 4$, the meets of size 2 are

$$\begin{aligned} A \wedge B \\ A \wedge C \\ B \wedge C \\ A \wedge D \\ B \wedge D \\ C \wedge D \end{aligned}$$

After adjoining a new generator E , the meets obtained by adjoining E to each of these are

$$\begin{aligned} A \wedge B \wedge E \\ A \wedge C \wedge E \\ B \wedge C \wedge E \\ A \wedge D \wedge E \\ B \wedge D \wedge E \\ C \wedge D \wedge E \end{aligned}$$

These appear as the suffix of the 3-meets for $n = 5$:

$$\begin{aligned} A \wedge B \wedge C \\ A \wedge B \wedge D \\ A \wedge C \wedge D \\ B \wedge C \wedge D \\ A \wedge B \wedge E \\ A \wedge C \wedge E \\ B \wedge C \wedge E \\ A \wedge D \wedge E \\ B \wedge D \wedge E \\ C \wedge D \wedge E \end{aligned}$$

Thus the meets obtained by adjoining the new generator form a contiguous suffix of the ordering, and within this suffix they appear in exactly the same relative order as the original sequence.

This suffix embedding persists as n increases. When a new generator A_{n+1} is introduced, every meet in $FD(n)$ produces a corresponding meet obtained by adjoining A_{n+1} . In particular, each $(n - 2)$ -meet T gives rise to the meet $T' = T \wedge A_{n+1}$ in $FD(n + 1)$.

Every $(n - 2)$ -subset of the generators in $FD(n + 1)$ either omits A_{n+1} , in which case it already appears in $FD(n)$, or includes A_{n+1} , in which case it is uniquely of the form $T \wedge A_{n+1}$ for some $(n - 2)$ -meet T of $FD(n)$. Thus no new $(n - 2)$ -meets arise beyond these translations.

To state this precisely, we introduce the following notation. For each meet T , let $\mathcal{S}_T$ denote the set of all elements whose leading (i.e., lexicographically minimal) meet is T .

This structural behavior explains the doubling pattern observed in Table 2 and yields the following general result.

Proposition 4. *Let T be an $(n - 2)$ -meet in $FD(n)$, and let $T' = T \wedge A_{n+1}$. Then*

$$|\mathcal{S}_{T'}| = 2|\mathcal{S}_T|.$$

Proof. Write

$$T = s_1 \wedge s_2 \wedge \cdots \wedge s_{n-2},$$

where $s_1, \dots, s_{n-2}$ are distinct generators in $FD(n)$, and let X and Y be the two generators not appearing in T . Thus the generators are $s_1, \dots, s_{n-2}, X, Y$.

We recall that elements of $\mathcal{S}_T$ are joins of meets whose smallest term in the canonical ordering is T . Any additional meet appearing in such a join must be lexicographically larger than T , since T is the leading meet. It must also be incomparable with T , since the joins are taken over antichains.

We first determine the $(n-1)$ -meets compatible with T .

An $(n-1)$ -meet omits exactly one generator. Since T omits the two generators X and Y , a meet of size $n-1$ can be incomparable with T only if it contains both X and Y (so that neither meet contains the other) and omits exactly one generator s_i from T .

Thus the admissible meets are obtained from T by replacing one generator s_i with the pair X, Y . There are therefore $n-2$ such meets.

Now adjoin a new generator A_{n+1} to T and define

$$T' = T \wedge A_{n+1}.$$

By the suffix property described above, the meets obtained by adjoining A_{n+1} to earlier meets appear at the end of the lexicographic ordering. In particular, T' occupies within this suffix the position corresponding to that of T among the earlier meets.

Within the block of elements whose leading meet is T' , the lower structure is inherited from that of $\mathcal{S}_T$. Indeed, every meet appearing in an element of $\mathcal{S}_T$ produces a corresponding meet in $FD(n+1)$ simply by adjoining A_{n+1} .

We next determine the $(n-1)$ -meets incomparable with T' .

Such a meet must share exactly $n-2$ generators with T' . There are two possibilities.

Type 1 (inherited meets). For each admissible meet in $FD(n)$ obtained by replacing a generator s_i with X and Y , adjoining A_{n+1} yields a meet that remains incomparable with T' . There are again $n-2$ such meets.

Type 2 (new meet). In addition, the meet

$$M = T \wedge X \wedge Y$$

shares $n-2$ generators with T' , since it contains all generators of T but omits A_{n+1} . This meet is lexicographically larger than T' and therefore may appear in joins with it.

These two cases exhaust all possibilities.

Thus the admissible $(n-1)$ -meets above T' consist of the $n-2$ inherited meets together with the single additional meet M . The number of admissible $(n-1)$ -meets therefore increases from $n-2$ to

$n - 1$.

We now show that the elements with leading meet T' can be partitioned into two disjoint families, each in bijection with $\mathcal{S}_T$.

For $x \in \mathcal{S}_T$, define

$$\varphi(x) = x \wedge A_{n+1}, \quad \psi(x) = (x \wedge A_{n+1}) \vee M.$$

Since x has leading meet T , the meet $x \wedge A_{n+1}$ has leading meet T' . Moreover, the meet M is lexicographically larger than T' , so adjoining it does not change the leading meet. Hence both $\varphi(x)$ and $\psi(x)$ have leading meet T' .

The images of φ and ψ are disjoint, since every element of the form $\psi(x)$ contains the meet M while no element of the form $\varphi(x)$ does.

Both maps are injective. If

$$\varphi(x_1) = \varphi(x_2),$$

then

$$x_1 \wedge A_{n+1} = x_2 \wedge A_{n+1},$$

which implies $x_1 = x_2$. A similar argument shows that ψ is injective.

Finally, let y be any element of $FD(n+1)$ whose leading meet is T' .

If y does not contain M , then every meet appearing in y contains A_{n+1} . Removing A_{n+1} from each meet therefore produces an antichain corresponding to an element $x \in \mathcal{S}_T$, and

$$y = x \wedge A_{n+1}.$$

If y does contain M , removing M from the join leaves an element of the form $x \wedge A_{n+1}$ for some $x \in \mathcal{S}_T$, so

$$y = (x \wedge A_{n+1}) \vee M.$$

Thus every element with leading meet T' lies in the image of either φ or ψ . The elements with leading meet T' therefore form two disjoint copies of $\mathcal{S}_T$, and

$$|\mathcal{S}_{T'}| = 2|\mathcal{S}_T|.$$

□

7 Conclusion

We have explicitly enumerated the elements of $FD(n)$ for $n \leq 7$, and organized this data according to the leading generators of their canonical representations. The resulting tables provide a detailed record of the distribution of elements across generators, and make visible several regularities that are not apparent from the lattice definition alone.

A key feature of this approach is the deliberate imposition of an ordering on the generators. While this breaks the inherent symmetry of $FD(n)$, it allows the lattice to be decomposed into more tractable pieces. Within this framework, patterns such as the recurrence for $g_i(n)$ and the doubling behavior for meets of size $n - 2$ emerge naturally from the data.

The enumeration data for $n \leq 7$ is included as part of this work and may serve as a basis for further investigation. In particular, preliminary examination of meets of size $n - 3$ suggests that additional combinatorial constraints arise beyond those observed in the $n - 2$ case. Understanding these constraints, and determining whether similar organizing principles persist at deeper levels, remains an open direction for future research.

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